

Automated Depression Detection Using CNN and Transformer-Based Pre-Trained Language Models

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Abstract. The research focuses on the automation of depression detection through the integration of Convolutional Neural Networks (CNNs) and Transformer-based Pre-Trained Language models. Traditionally, depression assessment relied on comprehensive clinical interviews where psychologists analyzed responses to assess an individual's mental state. In this model, the replication of this process incorporates three modalities: word context, audio, and video. Through the fusion of these modalities, the prediction of an individual's mental health status is achieved. The deep learning model encompasses three distinct data modalities, each representing different degrees of depression experienced by the subject. The model assigns appropriate weights to each modality, generating output based on their collective influence. This research contributes to advancing depression monitoring by employing cutting-edge deep learning techniques to analyze multi-modal data, leveraging independent learning synergies for enhanced accuracy.

Keywords: Depression detection, Convolutional Neural Networks (CNN), Long Short Term Memory (LSTM), Bi-directional Long Short Term Memory (BiLSTM), Natural Language Processing (NLP), Support Vector Machine (SVM), Random Forest (RF), Machine Learning(ML)

1 Introduction

Throughout the annals of human history, the shadow of depression has lingered, casting its profound influence on individuals and societies across cultures and epochs. The narrative of depression, both as a clinical phenomenon and a societal concern, transcends temporal boundaries, weaving a complex tapestry of human experience from ancient civilizations to the modern era.

In ancient times, depression was often intertwined with philosophical and spiritual perspectives. Ancient Greek physicians, such as Hippocrates, recognized a condition termed "melancholia," a state of chronic sadness and despondency. In ancient Chinese medicine, conditions resembling modern-day depression were explained as disruptions in the body's essential energies. Throughout the medieval era, perceptions of depression became more complex. Theologians and intellectuals associated melancholy with spiritual and moral shortcomings, viewing it as a consequence of divine displeasure. The medieval understanding, however, was not limited to moralistic views; it also incorporated notions of imbalances in bodily humors.

The advent of modern psychiatry in the 19th century marked a pivotal shift in comprehending and categorizing depressive disorders. Depressive disorder was identified as a distinct clinical condition by the Diagnostic and Statistical Manual of Mental Disorders (DSM), contributing to the ongoing enhancement of the categorization of mood disorders. This delineation facilitated a more systematic approach to diagnosis and treatment. Depression extends its tendrils beyond the individual psyche, leaving an indelible mark on relationships, work, and societal dynamics. The ripple effect of untreated depression manifests in diminished productivity, strained interpersonal relationships, and economic burdens. The societal toll of depression underscores the urgency for effective detection and intervention strategies.

Current approaches to depression treatment encompass a multifaceted toolkit, encompassing psychotherapy, pharmacotherapy, lifestyle modifications, and support networks. Despite significant progress, identifying and assisting individuals silently grappling with depression remains a challenging task. Against this backdrop, technological advancements offer a ray of hope. Automated Depression Detection, leveraging cutting-edge technologies such as Convolutional Neural Networks (CNNs) and Transformer-based pre-trained language models, present a promising frontier.

These models, trained on a vast corpus of textual, audio and visual data, can identify subtle patterns and cues indicative of depression, transcending the limitations of traditional diagnostic approaches. Facial expressions, audio, and textual data are synthesized to form a comprehensive profile, potentially enhancing the accuracy and efficiency of depression detection. The independent learning approach, combining the strengths of all three modalities, further amplifies the diagnostic capabilities, representing a significant change in our approach to evaluating mental health.

As we navigate the complex terrain of depression, the integration of automated detection tools serves as a beacon for early intervention and support. By expediting the diagnostic process, these technologies pave the way for timely and targeted interventions, thereby empowering individuals to reclaim their mental well-being. The fusion of ancient insights, modern understanding, and technology propels humanity to reveal new chapters in the ongoing saga of mental health.

2 Literature Review

The paper by Sayanta Paul, Jandhyala Sree Kalyani, and Tanmay Basu, titled "Early Identification of Anorexia and Depression Symptoms on Social Media with Efficient Machine Learning Fabrics," addresses challenges like early detection of depression and anorexia symptoms within dynamic social platforms, utilizing the CLEF eRisk 2018 dataset. The study employs classifiers to assess the efficacy of bag-of-words and UMLS features in identifying depression, with the bag of words yielding the most favorable outcomes. The limited efficacy of RNN models is attributed to their constrained performance.[1]

A strategy to optimize word embeddings for parentheses is proposed in the paper "Deep Learning for Depression Detection of Twitter Druggies" authored by Ahmed Hussein Orabi, Mahmoud Hussein Orabi, Diana Inkpen, and Prasadith Buddhitha. The utilization of CNN and RNN methods to the CLPsych2015 and Bell Lets Talk datasets revealed that models anchored in CNN exhibited superior performance compared to those grounded in RNN.[2]

Ji, C.P. Yu, S.-F. Fung, S. Pan, and G. Long, "Supervised literacy for suicidal creativity discovery in online stoner content," uses two datasets, from Reddit and Twitter. This identifies features from colorful perspectives and makes use of traditional and deep literacy classifiers to compare similar as Random Forest, Gradient Boost Bracket tree, XGBoost, MLFFNN and LSTM out of which XGBoost has the loftiest AUC of 0.9569. [3]

Ji, X. Li, Z. Huang, and E. Cambria, "Suicidal creativity and internal complaint discovery with attentive relation networks," focuses on enabling relational textbook garbling to classify suicidal creativity and internal diseases. The suggested system is evaluated against the birth models, fastText, CNN, LSTM, RCNN and SSA enforced by PyTorch.[4]

Zhang, T., Schoene, A.M., Ji, S. et al. Natural language processing applied to internal illness discovery a narrative review uses NLP styles and Deep literacy styles to dissect data of which the ultimate has shown better performance. This is mostly because Deep Learning methods may automatically extract valuable features without requiring point engineering.[5]

Margaret Mitchell, Kristy Hollinshead, and Glen Coppersmith. Quantifying the linguistic patterns of Schizophrenia in Social Media, attempts to define features applicable to internal health and schizophrenia and explore the differences between schizophrenia exhibiting negative and positive druggies regarding the specified features. Automatic NLP styles are used and patterns are linked.[6]

Using colorful NLP styles in conjunction to determine which pairing yields a higher vaticination delicacy, Luna Ansari, Shaoxiong Ji, Qian Chen, and Erik Cambria's Ensemble Mongrel Learning styles for Automated Depression Discovery performs depression discovery across three datasets.[7]

In the work by Mukherjee, S.S. et al. titled "Natural language processing-grounded quantification of the internal state of psychiatric cases" (Comput. Psychiatry 4, 76 – 106, 2020), a dual vector representation of symptoms and emotional states in psychiatric cases is generated employing deep learning and NLP techniques grounded in the individual's Subjective Assessment (SA). The goal is to formulate new treatment plans by comparing the challenges faced by patients with their internal health conditions.[8]

Gong et. al. uses a regression method based on PLS wherein a late fusion detection method along with the question-answer based model, applying it on all 3 modalities for model prediction.[9]

3 Existing System

The user inputs various features, which are subsequently parameterized and analyzed through diverse methods. Additionally, studies have shown that Natural Language Processing (NLP) techniques deliver excellent results in deep learning applications, particularly in text categorization and image recognition. Recurrent Neural Networks (RNNs) constitute a group of neural network architectures where the output of the current segment is directly connected to its own input.

Word processing also frequently makes use of LSTM and Bi-Directional LSTM, the latter of which shortens processing times. The construction of most models is typically based on a predefined set of lexicons, or a prefixed dictionary of data, acting as a data source from which the model may compare and determine features. Sentiment representations of sentences, including polarity label, sensitivity, and value, are components of the widely used lexicon SenticNet.

The binary classifiers obtained from the preprocessing phases are then trained utilizing machine learning techniques, and the outcomes are assessed for accuracy and other pertinent metrics.

4 Proposed System

Traditionally, depression detection relied on thorough clinical interviews, during which psychologists analyzed the subject's responses to assess their mental state. In the model presented, an attempt is crafted to emulate this approach by integrating three modalities such as word context, audio, and video to predict the patient's mental health. The output is stratified into different levels, considering the varying degrees of depression in the subject. A model has been constructed to fuse these modalities, assigning suitable weights to each and thereby generating an output.

4.1 Dataset

The University of Southern California collected the DAIC-WOZ dataset, a portion of the broader DAIC (Distress Analysis Interview Corpus), designed to aid in diagnosing psychological distress conditions like anxiety, depression, and PTSD.

Comprising audio and video recordings along with extensive questionnaire responses, the DAIC-WOZ dataset incorporates Wizard-Of-Oz interviews. These sessions involve an animated virtual assistant named Ellie, controlled by a human interviewer in a separate room. The data is transcribed and annotated to capture various verbal and non-verbal features. Each participant's session includes a transcription of the interaction, participant audio files, and facial features derived from the recorded video.

In total, the dataset encompasses 189 interaction sessions, varying in duration from 7 to 33 minutes. It includes interviews with 59 individuals experiencing depression and 130 non-depressed subjects.

4.1 Modalities

4.1.1 Audio Modality

The attributes are sampled at a rate of 100Hz with a granularity of 10 milliseconds. The feature set includes 12 Mel-frequency cepstral coefficients (MFCCs), namely FO, QOQ, VUV, NAQ, H1H2, PSP, MDQ, Rdconf, Rd, MCEPO-24, peakSlope, HMPDM0-24, and HMPDD0-12. Complementary to the MFCCs, there are supplementary attributes for pitch tracking, peak slope, maximal dispersion quotients, and glottal source parameters. The VUV feature indicates whether the current sample is voiced or unvoiced, and in cases where the sample is unvoiced (VUV = 0), FO, QOQ, NAQ, MDQ, H1H2, PSP, Rd, and peakSlope are set to 0.

4.1.2 Video Modality

The dataset encompasses various facial attributes obtained from participant videos. These include 68 2D points on the face, 24 Action Units (AU) features measuring facial activity, 68 3D points on the face, 16 features representing the subject's gaze, and 10 features reflecting the subject's pose. In total, this results in 388 distinct video features.

4.1.3 Text Modality

Within the textual modality, the dataset contains the entire conversation transcript between the patient in CSV format. Individual sentences are time stamped and classified based on their speaker. Expressions such as laughter or frowns are encapsulated in angular brackets (e.g., <Laughter>) whenever they occur. It is worth mentioning that the differentiation between long and short pauses has not been made, and only word-level segmentation, not phoneme-level, has been recorded.

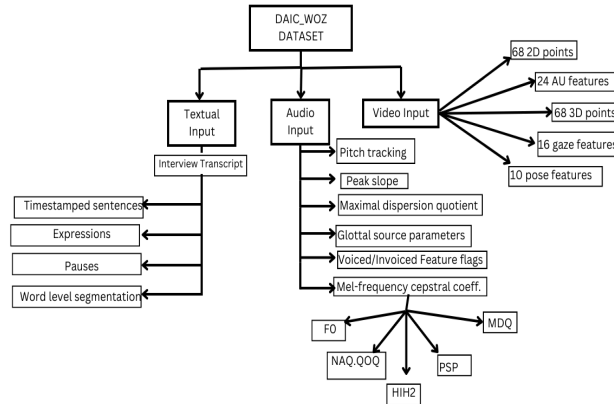


Figure 1. Features Used in DAIC_WOZ Dataset

4.3 Model Flow

In the proposed system, the approach involves initial feature extraction, followed by the implementation of a gating mechanism and hybrid fusion technique on the extracted features. The features consist of audio, visual, and textual modalities, integrated using timestamps to capture time-dependent interactions. Sentence-level forced alignment is conducted during preprocessing to enable the framework to learn contextual relationships between words.

Having aligned the textual, visual and audio features at the sentence level, it's crucial to acknowledge that different modalities may have varying impacts on the final result, introducing noise to the representation of features. To address this, a gating mechanism is applied to the derived features to regulate the influence of each modality to the final output. Weight vectors linked to each modality are employed in the network to control and learn the transformation and information carried to subsequent layers.

At each time interval, feature representations from every modality are merged then passed through a word-level LSTM containing the gating mechanism. Prior to the concatenation of feature vectors, gating mechanisms are also applied to the visual, textual and audio vectors to extract essential information.

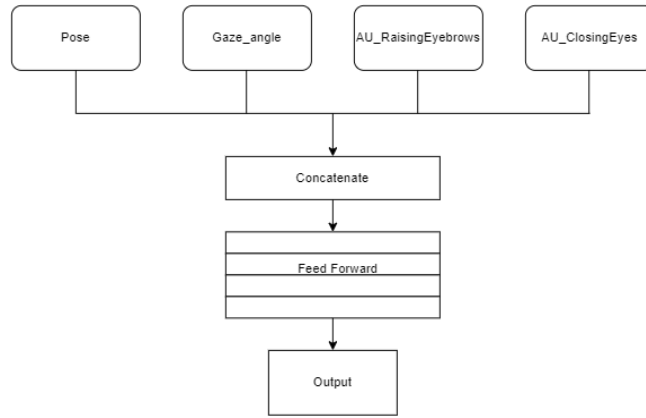


Figure 2. Late Fusion Model Architecture

4.4 Training Procedure

4.4.1 SVM and Random Forest

Initially, SVM (utilizing an RBF kernel) and Random Forest were individually utilized on the three modalities. Subsequently, another SVM model underwent training on the decision labels obtained from each specific modalities to facilitate late fusion. In this context, the features of the video and audio modalities were averaged across all timestamps, resulting in 74 and 388 features, respectively. For the text modality, the word2Vec model from google-news-300 was employed to transform each word into a 300-dimensional vector. The resulting 3D vector (sentences x words x 300 features) was initially averaged over each word and then flattened.

4.4.2 CNN

A 6-layer CNN model was constructed, with the initial four layers incorporating conv2D layers for the text modality and conv1D layers for audio and video modalities, along with Max Pooling layers. Additional flattening and fully connected layers were integrated with ReLU activation, while the last layer utilized Sigmoid activation. For audio and video modalities, attributes were derived from the first 40,000 timestamps, chosen based on computational capabilities. Concerning the text modality, after applying the Word2vec model, thresholds were set for the maximum number of sentences and words.

4.4.3 LSTM Model with/without Gating (Sentence-level)

Initially, the data underwent transformation to the sentence level by averaging features from the audio and video modalities corresponding to the provided timestamps of sentences. This involved the application of sentence-level forced alignment.

4.4.4 (Bi)LSTM Model with Gating (Word-Level)

The data underwent forced alignment at a word level. This alignment employed a unitary approach, converting sentence-level timestamps to word-level depending on the count of words in the sentence and the character length present in each sentence.

Model	Feature	F1	Prec	MAE	RMSE
Baseline					
DAIC Baseline	Audio+Video	-	-	5.66	7.05
Gong et al.	Text+Audio+Video	0.60	-	3.96	4.99
Alhanai et al.	Text	0.66	0.70	5.09	6.11
Alhanai et al.	Text+Audio	0.75	0.72	5.02	6.04
Williamson et al.	Text	0.67	0.74	3.82	5.06
Williamson et al.	Text+Audio+Video	0.70	0.78	3.84	5.23
Word Level Models					
LSTM	Text	0.68	0.69	4.99	6.04
LSTM	Text+Audio	0.66	0.69	5.19	6.41
LSTM	Text+Audio+Video	0.66	0.64	5.28	6.69
LSYM with Gating	Text+Audio	0.81	0.77	3.67	5.15
LSTM with Gating	Text+Audio+Video	0.81	0.80	3.61	4.99

Table 2. Baseline Results

Following this, the outcomes are combined with the corresponding text feature, and the resulting concatenated vector undergoes further processing through a (Bi)LSTM layer to produce the final output.

In more detail, the audio and video features initially pass through three feedforward highway layers. Subsequently, dense layers are employed to diminish the dimensionality of both video and text features.

After concatenation, a (Bi)LSTM with 128 hidden nodes is applied, and a last compact layer with sigmoid activation generates the results. A learning rate of 0.0001 is set, and the iteration count is controlled using the EarlyStopping callback from the Keras API.

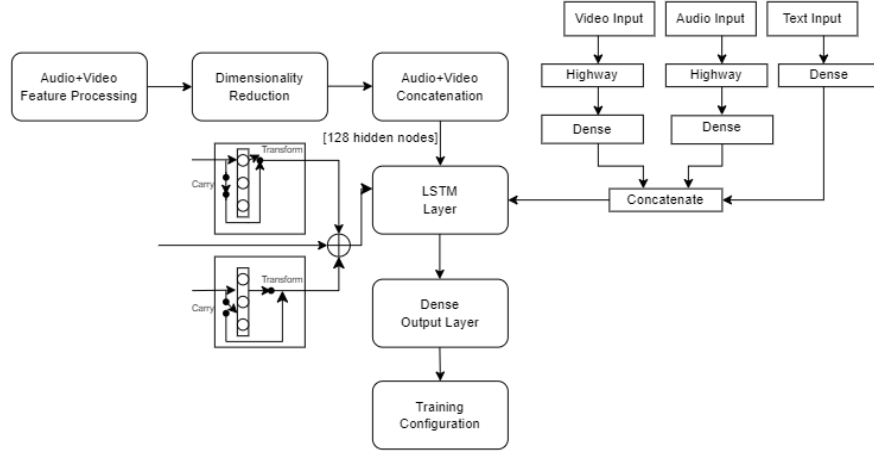


Figure 3. Model Architecture Using Sentence-Level Multi-Modal Fusion With Gating

5 Results and Evaluation

The outcomes were reported by computing a weighted mean for two classes, namely class 0 (Not depressed) and class 1 (Depressed). The data distribution is maintained at a ratio of 7:3.

5.1 SVM Model

The SVM model demonstrated less-than-satisfactory performance, as indicated by the results. This suboptimal outcome may stem from the utilization of averaging operations across the three modalities, potentially causing information loss and resulting in the model's underperformance.

5.2 CNN Model

Unlike the SVM model, the CNN model exhibited better results in the text modality because of the absence of averaging across word vectors. However, the audio and video modalities still yielded unsatisfactory outcomes, likely attributable to the limited data points and sparse features characterizing these modalities.

5.3 LSTM with/without Gating at Sentence-Level

The results unequivocally highlight the superior performance of the framework in the text modality. The relatively low F1 score values in the video and audio modalities suggest that these features inadequately represent the depression class. Consequently, when combining video, audio and text modalities, the results fall just short of those achieved by the model solely utilizing the text modality.

Moreover, models incorporating gating mechanisms consistently outperform those without gating. This could be ascribed to the reality that gating selectively amplifies the most crucial features while nullifying others, facilitating a form of feature extraction at this level. This, in turn, aids our LSTM model in learning from only the most influential features.

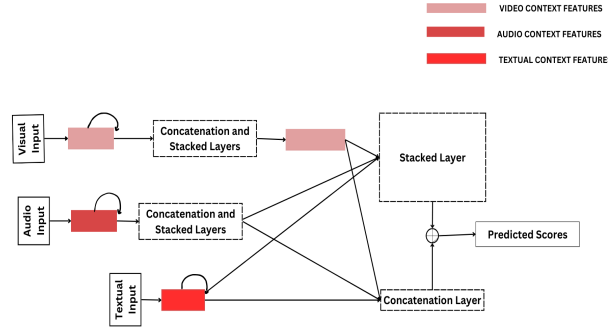


Figure 4. Block Diagram of Proposed Network on Multi-Modality Input Feature

5.4 LSTM with gating at word-level

The outcomes for the word-level LSTM are not as favorable as anticipated. This might be assigned to the model's limited contextual understanding of the conversation at the word level in contrast to the sentence level.

5.5 BiLSTM Model

This model exhibits a noticeable inclination towards the depressed class, indicating a challenge in effectively learning from the given data.

Model	Modality	Precision	Recall	F1-Score
Late Fusion Using SVM	Text+Audio+ Video	0.443	0.386	0.423
CNN	Text	0.567	0.627	0.597
	Audio	0.086	0.4	0.134
	Video	0.086	0.4	0.136
LSTM	Text	0.656	0.67	0.667
	Audio	0.575	0.454	0.456
	Video	0.50	0.676	0.568
BiLSTM (word_Level)	Text+Audio+ Video	0.78	0.322	0.188

LSTM w/o Gating (Sentence- Level)	Text+Audi o	0.713	0.358	0.475
	Text+Vide o	0.605	0.468	0.475
	Text+Audi o+Video	0.622	0.489	0.6
LSTM w Gating (Sentence_Level)	Text+Audi o	0.634	0.653	0.647
	Text+Vide o	0.652	0.538	0.559
	Text+Audi o+Video	0.639	0.653	0.647
LSTM w Gating (Word_Level)	Text+Audi o+Video	0.490	0.652	0.554

Table 2. Precision, Recall, F1-Score of Used Models

6 Conclusion

In conclusion, the amalgamation of Transformer-based pretrained language models and CNN in the presented model for automated depression detection shows promise in enhancing accuracy and robustness. The model, utilizing indicators from text, video, and audio modalities, employs a sentence-level approach with highway layers as a gating mechanism. Our f

Findings suggest that the sentence-level model outperforms other models, and a combination of both early and late fusion improves modality interpretation.

For future research, a more comprehensive extraction of features, including additional audio features like response time and pauses, could offer a clearer insight of symptoms. Exploring the correlation amid bodily action sequences from motion capture data and verbal behavior could also contribute to a more comprehensive study.

Regular evaluation on a validation set, along with potential hyperparameter fine-tuning, ensures adaptability to diverse dataset characteristics.

The model demonstrates potential for enhanced accuracy and robustness in addressing the challenging task of automated depression identification, leveraging a synergistic combination of conventional machine learning and deep learning features.

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